

# Yantong (Kelly) Chen

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## Education

### Emory University

Atlanta, United States

*Master of Science in Public Health: Biostatistics*

08/2023 – 05/2025

*Key Coursework:* Statistical Inference, Applied Linear Model, Probability Theory, Epidemiologic Methods, Clinical Trial, Machine Learning, Survival Analysis Methods, Modern Regression Analysis

### University of British Columbia

Vancouver, Canada

*Bachelor of Science: Mathematics*

09/2019 – 05/2023

*Key Coursework:* Integral Calculus, Matrix Algebra, Probability, Applied Linear Algebra, Algebra and Coding Theory, Statistical Inference, Linear Programming, Graphs and Networks

## Technical Skills

**Programming Languages:** Python (5-year), SAS (4-year), R (4-year), MATLAB (2.5-year), JavaScript (1-year)

## Professional Experience

### Research Data Analyst

Boston, MA

*Slone Epidemiology Center, Boston University*

09/02/2025 – Present

- Conducted RNA-seq analyses in breast cancer cases from the Black Women's Health Study (BWHS) to investigate gene expression, pathway, and immune-profile differences associated with obesity, type 2 diabetes, and antihypertensive medication use. Applied DESeq2 differential expression, Reactome GSEA, ssGSEA, and CIBERSORTx, with stratified analyses by ER and molecular characteristics and extensive evaluation of sequencing batch effects using R/Bioconductor and shell scripting.
- Investigated associations between PAM50 molecular subtypes and breast cancer tumor characteristics by integrating PAM50 classification, P53 mutation status, CD8<sup>+</sup> staining, pathology, and epidemiologic data; performed immune-subtype classification and stratified association analyses across ER and tumor subgroups.
- Developed a spatial tissue microarray analysis workflow to quantify and visualize immune and tumor-cell markers across epithelial and stromal compartments and evaluate their associations with epidemiologic factors and breast cancer characteristics.
- Managed and harmonized large-scale BWHS epidemiologic questionnaire, RNA-seq, pathology, and tissue microarray datasets and fulfilled data requests supporting multiple collaborative breast cancer research projects.
- Developed an interactive R Shiny data exploration platform for the BWHS database, enabling research team members to examine variable distributions, summary statistics, participant-level and longitudinal information, and download customized data subsets for downstream analyses.

### Special Volunteer

*Division of Cancer Epidemiology and Genetics, National Cancer Institute*

02/2025 – Present

- Contributing as part of a 10+ member research team to a large-scale scoping review of 600+ ovarian cancer-related publications, synthesizing evidence on genetic and epidemiologic risk factors.
- Analyzing data from the Prostate, Lung, Colorectal and Ovarian (PLCO) Cancer Screening Trial to integrate polygenic risk scores (PRS) with epidemiologic variables, evaluating improvements in ovarian cancer risk stratification beyond traditional risk factors.

### Research Assistant Intern

Atlanta, GA

*American Cancer Society*

08/19/2024 – 02/19/2025

- Analyzed polygenic risk scores (PRS) for ovarian cancer to evaluate whether genetic risk prediction improves absolute risk estimation beyond established epidemiologic factors within the Cancer Prevention Study II (CPS-II) cohort.
- Utilized R programming to analyze genomic data and implement statistical models for cancer epidemiology.
- Applied the iCare package to develop and validate PRS models, ensuring rigorous evaluation of genetic risk factors.

**Graduate Research Assistant**  
*Emory School of Medicine*

Atlanta, GA  
07/18/2024 – 05/01/2025

- Analyzed literacy, gender, education, and stress for 191 participants using SAS to study aging perceptions in the South Asian community.
- Cleaned data and conducted survival and mixed model analyses for 1,253 participants to explore the link between pulse wave velocity and white matter hyperintensity in Alzheimer's research using SAS.

**Nursing Administration Intern**  
*UCLA Health*

Los Angeles, CA  
06/10 – 08/16/2024

- Standardized Scope of Services: Reviewed and summarized services for 33 clinics, implemented a new platform for form submissions, and developed a quantifying supervision approach using data analysis skills.
- Reduced Medication Retrieval Time: Observed workflows during pharmacy shifts, developed strategies to reduce nurse travel time, created a process map, and identified key improvement areas.
- Participated in Lean Academy studies, HIC training, facility tours, executive meet-and-greets, and the Journal Club, enhancing knowledge in healthcare operations and management.

**Graduate Research Assistant**  
*Emory School of Nursing*

Atlanta, GA  
01/28 – 06/08/2024

- Analyzed ECG waveform data from Grady Hospital using MATLAB to develop AI tools for emergency cardiac care.
- Collaborated with Duke University on cancer research, assessing symptom improvement and intervention effectiveness in at-home cancer patients using Python for statistical analysis.

## Research Experience

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**Integrative Statistical Methods for Environmental Exposure Assessment** 10/2023 – 06/01/2025

- Conducted independent research on integrating Quantitative G-Computation (qgcomp) and Bayesian Additive Regression Trees (BART) using R to analyze environmental exposure data.
- Developing a novel statistical framework to improve the accuracy and interpretability of health impact assessments from complex environmental exposures.

**Adolescent Mental Health Empowerment Initiative** 01/11 – 04/26/2024

- Contributed to a team project selected as a finalist in the Interprofessional Education Project Program, focusing on a comprehensive approach to adolescent mental health.
- Developed strategy integrating Prevention, Intervention, and Connection to provide early education, targeted screenings, and direct linkage to mental health resources.

**Analysis of Drawings of Parkinson's Patients Based on Machine Learning** 11/2022 – 04/2023

- Collected and analyzed drawings from healthy individuals and Parkinson's patients across varying ages, genders, and races using both dominant and non-dominant hands to draw periodic ring-shaped circles.
- Identified and classified Parkinson's disease in painters through machine learning techniques.

## Academic Experience

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**Bruce Weir Summer Institute in Statistical Genetics (SISG)** 05/29 – 06/07/2024

- *Genetics and Genomics*: Studied heritability, inheritance, genetic variants, genome sequencing, gene expression profiling, and evolutionary factors.
- *Bayesian Statistics*: Learned Bayesian analysis principles, applied MCMC and INLA for genetic data, and used Stan, JAGS, INLA, and WinBUGS in R.
- *Statistical Genetics*: Estimated allele frequencies, inbreeding coefficients, Hardy-Weinberg equilibrium, kinship levels, and population structure using R packages including *gaston*, *hierfstat*, *HardyWeinberg*, and *adegenet*.

- Conducted a detailed analysis of South Africa's health system performance, epidemiologic transitions, and social health determinants as part of a global health course.
- Collaborated with classmates to design a capstone intervention focused on enhancing political leadership in the implementation of National Health Insurance.

- Analyzed surgical site infection (SSI) data from the CDC National Healthcare Safety Network (NHSN).
- Conducted statistical analysis to identify SSI trends and risk factors for various operative procedures.
- Utilized Python and Excel for data cleaning, visualization, and statistical modeling.

## **Publication**

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Albrecht T, Chen Y, Allen D, Pastva A, Tian Z, Lin Y, Bold D, Hu X, Xiao R. Digital Health Innovations to Promote Physical Activities in Patients with Cancer. *Stud Health Technol Inform*. 2024 Jul 24;315:608–609. doi: 10.3233/SHTI240239. PMID: 39049345.